



Qualification Pack



Bio Gas Plant & Bio Slurry Technician

QP Code: WBSC/SGJ/Q0201

Version: 1.0

NSQF Level: 3

WB State Council of Technical and Vocational Education and Skill Dev || 5th Floor, Karigori Bhawan,
Plot- b/7, Action Area-III, New Town
Rajarhat || email:pinaki@wbscte.ac.in



Qualification Pack

Contents

WBSC/SGJ/Q0201: Bio Gas Plant & Bio Slurry Technician	3
<i>Brief Job Description</i>	3
Applicable National Occupational Standards (NOS)	3
<i>Compulsory NOS</i>	3
<i>Qualification Pack (QP) Parameters</i>	3
WBSC/SGJ/N0201: Identify Biomass resource and Bio-energy potential.	5
WBSC/SGJ/N0202: Illustrate mechanism of Biogas Systems.	9
WBSC/SGJ/N0203: Explain the process of Biomass Gasifiers.	13
WBSC/SGJ/N0204: Identify appropriate site for installing the BGP.	17
WBSC/SGJ/N0205: Construct a Bio Gas Plant with proper care and safety.	21
WBSC/SGJ/N0206: Install Cooking Apparatus.	27
WBSC/SGJ/N0207: Carry out post BGP construction activities.	30
WBSC/SGJ/N0208: Demonstrate the procedure of Bio slurry manure making and Marketing of Organically grown crops.	34
WBSC/SGJ/N0209: Work in real job situation with special emphasis on basic safety and hazards in this domain (Bio Gas Plant & Bio Slurry Technician).	38
DGT/VSQ/N0102: Employability Skills (60 Hours)	41
Assessment Guidelines and Weightage	48
<i>Assessment Guidelines</i>	48
<i>Assessment Weightage</i>	49
Acronyms	51
Glossary	52



Qualification Pack

WBSC/SGJ/Q0201: Bio Gas Plant & Bio Slurry Technician

Brief Job Description

A Bio Gas Plant & Bio Slurry Technician is responsible for the construction, operation and maintenance of bio gas plants and the efficient management of bio-slurry for agricultural applications. The role involves applying professional technical knowledge and practical skills to produce clean energy, manage organic waste and promote sustainable farming practices. Technicians support farmers by reducing input costs, improving soil fertility and enhancing productivity through the use of bio-slurry as an organic fertilizer.

Personal Attributes

A Bio Gas Plant & Bio Slurry Technician should possess a strong interest in sustainable agriculture, clean energy and environmental protection along with a practical and problem-solving mindset. The individual should be technically inclined, safety-conscious and capable of working both independently and in rural field conditions. Good communication skills, teamwork and a sense of social responsibility are important as the technician often works closely with farmers, rural youth and local institutions to promote eco-friendly practices, enhance livelihoods and contribute to national sustainable development goals.

Applicable National Occupational Standards (NOS)

Compulsory NOS:

1. [WBSC/SGJ/N0201: Identify Biomass resource and Bio-energy potential.](#)
2. [WBSC/SGJ/N0202: Illustrate mechanism of Biogas Systems.](#)
3. [WBSC/SGJ/N0203: Explain the process of Biomass Gasifiers.](#)
4. [WBSC/SGJ/N0204: Identify appropriate site for installing the BGP.](#)
5. [WBSC/SGJ/N0205: Construct a Bio Gas Plant with proper care and safety.](#)
6. [WBSC/SGJ/N0206: Install Cooking Apparatus.](#)
7. [WBSC/SGJ/N0207: Carry out post BGP construction activities.](#)
8. [WBSC/SGJ/N0208: Demonstrate the procedure of Bio slurry manure making and Marketing of Organically grown crops.](#)
9. [WBSC/SGJ/N0209: Work in real job situation with special emphasis on basic safety and hazards in this domain \(Bio Gas Plant & Bio Slurry Technician\).](#)
10. [DGT/VSQ/N0102: Employability Skills \(60 Hours\)](#)



Qualification Pack

Qualification Pack (QP) Parameters

Sector	Green Jobs
Sub-Sector	Waste Management
Occupation	Bio Gas Plant & Bio Slurry Technician
Country	India
NSQF Level	3
Credits	15
Aligned to NCO/ISCO/ISIC Code	3142.9900
Minimum Educational Qualification & Experience	10th Class OR 8th grade pass and pursuing continuous schooling in regular school with vocational subject OR 8th grade pass with 2 Years of experience OR 5th grade pass with 5 Years of experience
Minimum Level of Education for Training in School	
Pre-Requisite License or Training	NA
Minimum Job Entry Age	18 Years
Last Reviewed On	NA
Next Review Date	14/12/2026
NSQF Approval Date	14/12/2023
Version	1.0
Reference code on NQR	QG-03-ES-00362-2023-V2-WBSC
NQR Version	2



Qualification Pack

WBSC/SGJ/N0201: Identify Biomass resource and Bio-energy potential.

Description

Identify Biomass resource and Bio-energy potential.

Scope

The scope covers the following :

- Classification of biomass.
- Bio Energy Resources, World Bio Energy Potential, Bio Energy Potential of India.
- Properties and characteristics of biomass.
- Different energy conversion methods.
- Biomass Combustion methods.
- Utilization of bioenergy in Biomass based power generation and domestic cooking.

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

PC1. Classify biomass.

PC2. Identify Bio Energy Resources, World Bio Energy Potential, India's Bio Energy Potential.

PC3. Identify properties and characteristics of biomass.

PC4. Identify different energy conversion methods.

PC5. Identify Biomass Combustion methods.

PC6. Utilize bioenergy in Biomass based power generation and domestic cooking.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. Source biomass and bioenergy.

KU2. Usage of appropriate biomass for the generation of bioenergy.

KU3. Physio-chemical characteristics of biomass.

KU4. Different energy conversion methods.

KU5. Biomass Combustion methods.

KU6. Usage of bioenergy in Biomass based power generation and domestic cooking.

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. Identify Bio Energy Resources.

GS2. Select suitable biomass for generation of bio energy.

GS3. Check the availability of biomass.



Qualification Pack

- GS4.** Demonstrate Physio-chemical characteristics of biomass.
- GS5.** Verify Biomass Combustion potential and Loose biomass densification capacity.
- GS6.** Find Biomass based power generation and domestic cooking process.



Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	10	80	-	-
PC1. Classify biomass.	-	-	-	-
PC2. Identify Bio Energy Resources, World Bio Energy Potential, India's Bio Energy Potential.	-	-	-	-
PC3. Identify properties and characteristics of biomass.	-	-	-	-
PC4. Identify different energy conversion methods.	-	-	-	-
PC5. Identify Biomass Combustion methods.	-	-	-	-
PC6. Utilize bioenergy in Biomass based power generation and domestic cooking.	-	-	-	-
NOS Total	10	80	-	-



Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	WBSC/SGJ/N0201
NOS Name	Identify Biomass resource and Bio-energy potential.
Sector	Green Jobs
Sub-Sector	
Occupation	Bio Gas Plant & Bio Slurry Technician
NSQF Level	3
Credits	1
Version	1.0
Last Reviewed Date	14/12/2023
Next Review Date	14/12/2026
NSQC Clearance Date	14/12/2023



Qualification Pack

WBSC/SGJ/N0202: Illustrate mechanism of Biogas Systems.

Description

Illustrate mechanism of Biogas Systems.

Scope

The scope covers the following :

- The Technology of Biogas production, and types of Digester.
- Bio methanation process and Chemical reaction of bio methanation process.
- Design of Digester, Biogas plant design, operation, fault checking, maintenance.
- The use of Dung, Vegetable Waste and Municipal Waste based Biogas plants.
- The procedure of Biogas Bottling Plant Technology.
- Utilization of Biogas as fuel for transportation, Lighting, Running Dual Fuel Engines, Electricity generation.

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

- PC1.** Describe the Technology of Biogas production, and Digester types.
- PC2.** Exhibit Bio methanation process and Chemical reaction of bio methanation process.
- PC3.** Demonstrate design of Digester, Biogas plant design, operation, fault checking, maintenance.
- PC4.** Explain the use of Dung, Vegetable Waste and Municipal Waste based Biogas plants.
- PC5.** Demonstrate the procedure of Biogas Bottling Plant Technology.
- PC6.** Utilize Biogas as fuel for transportation, Lighting, Running Dual Fuel Engines, Electricity generation.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Introduction to bio gas.
- KU2.** Difference between Biogas and LPG/CNG.
- KU3.** Advantages of Biogas.
- KU4.** Bio methanation process and Chemical reaction of bio methanation process.
- KU5.** Criteria for design of Biogas plant.
- KU6.** Dung, Vegetable Waste and Municipal Waste based Biogas plants.
- KU7.** Availability of water in the vicinity for making a water solution of dung to be given in the input inlet.
- KU8.** Procedure of bottling of Biogas.
- KU9.** Methods of use of Biogas as fuel for transportation, Lighting, Running Dual Fuel Engines, Electricity generation.



Qualification Pack

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** Demonstrate Biogas generation process with the chart/videos.
- GS2.** Demonstrate Biogas plant design and Digester design.
- GS3.** Demonstrate various Chemical reaction involve in bio methanation process.
- GS4.** Demonstrate construction of Dung, Vegetable Waste and Municipal Waste based Biogas plants.
- GS5.** Demonstrate Biogas Bottling system and procedure.



Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	16	60	-	-
PC1. Describe the Technology of Biogas production, and Digester types.	16	-	-	-
PC2. Exhibit Bio methanation process and Chemical reaction of bio methanation process.	-	-	-	-
PC3. Demonstrate design of Digester, Biogas plant design, operation, fault checking, maintenance.	-	-	-	-
PC4. Explain the use of Dung, Vegetable Waste and Municipal Waste based Biogas plants.	-	-	-	-
PC5. Demonstrate the procedure of Biogas Bottling Plant Technology.	-	-	-	-
PC6. Utilize Biogas as fuel for transportation, Lighting, Running Dual Fuel Engines, Electricity generation.	-	-	-	-
NOS Total	16	60	-	-



Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	WBSC/SGJ/N0202
NOS Name	Illustrate mechanism of Biogas Systems.
Sector	Green Jobs
Sub-Sector	
Occupation	Bio Gas Plant & Bio Slurry Technician
NSQF Level	3
Credits	1
Version	1.0
Last Reviewed Date	14/12/2023
Next Review Date	14/12/2026
NSQC Clearance Date	14/12/2023



Qualification Pack

WBSC/SGJ/N0203: Explain the process of Biomass Gasifiers.

Description

Explain the process of Biomass Gasifiers.

Scope

The scope covers the following :

- The Principle of Bio mass Gasifiers.
- Design of Bio mass Gasifiers, updraft gasifier, down draft gasifier.
- Zero carbon biomass gasification plants.
- Gasification of plastic-rich waste.
- Biomass integrated gasification/combined cycles systems, gasification, pyrolysis, liquefaction process.
- Biomass pre- treatment and processing.
- Electricity generation from biomass gasifier, bio-gasoline, bio-diesel and duel fuel engine.

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

- PC1.** Understand the Principle of Bio mass Gasifiers.
- PC2.** Interpret Design of Bio mass Gasifiers, updraft gasifier, down draft gasifier.
- PC3.** Demonstrate Zero carbon biomass gasification plants.
- PC4.** Demonstrate Gasification of plastic-rich waste.
- PC5.** Understand Biomass integrated gasification/combined cycles systems, gasification, pyrolysis, liquefaction process.
- PC6.** Demonstrate Biomass pre- treatment and processing.
- PC7.** Demonstrate Electricity generation from biomass gasifier, bio-gasoline, bio-diesel and duel fuel engine.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** The Design of Bio mass Gasifiers, updraft gasifier, down draft gasifier.
- KU2.** Construction of Zero carbon biomass gasification plants.
- KU3.** Gasification of plastic- rich waste.
- KU4.** Biomass integrated gasification, pyrolysis, liquification process.
- KU5.** Biomass pre- treatment and processing.
- KU6.** Electricity generation from biomass gasifier, bio-gasoline, bio- diesel and duel fuel engine.

Generic Skills (GS)



Qualification Pack

User/individual on the job needs to know how to:

- GS1.** Demonstrate Design of Bio mass Gasifiers , updraft gasifier, down draft gasifier.
- GS2.** Demonstrate construction of Zero carbon biomass gasification plants.
- GS3.** Construct gasifier for Gasification of plastic-rich waste.
- GS4.** Demonstrate Biomass integrated gasification cycles systems, gasification, pyrolysis, liquification cycles.
- GS5.** Demonstrate Biomass pre- treatment and processing.
- GS6.** Demonstrate biodiesel, improved biomass cookstove, biohydrogen generation procedure with the chart/videos.
- GS7.** Demonstrate Electricity generation process from biomass gasifier, engine systems, bio-gasoline, bio-diesel and duel fuel engine with the chart/videos.



Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	20	60	-	-
PC1. Understand the Principle of Bio mass Gasifiers.	-	-	-	-
PC2. Interpret Design of Bio mass Gasifiers, updraft gasifier, down draft gasifier.	-	-	-	-
PC3. Demonstrate Zero carbon biomass gasification plants.	-	-	-	-
PC4. Demonstrate Gasification of plastic-rich waste.	-	-	-	-
PC5. Understand Biomass integrated gasification/combined cycles systems, gasification, pyrolysis, liquefaction process.	-	-	-	-
PC6. Demonstrate Biomass pre- treatment and processing.	-	-	-	-
PC7. Demonstrate Electricity generation from biomass gasifier, bio-gasoline, bio-diesel and duel fuel engine.	-	-	-	-
NOS Total	20	60	-	-



Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	WBSC/SGJ/N0203
NOS Name	Explain the process of Biomass Gasifiers.
Sector	Green Jobs
Sub-Sector	
Occupation	Bio Gas Plant & Bio Slurry Technician
NSQF Level	3
Credits	1
Version	1.0
Last Reviewed Date	14/12/2023
Next Review Date	14/12/2026
NSQC Clearance Date	14/12/2023



Qualification Pack

WBSC/SGJ/N0204: Identify appropriate site for installing the BGP.

Description

Identify appropriate site for installing the BGP.

Scope

The scope covers the following :

- Selection of suitable location.
- Availability of water in the vicinity for making a water solution of dung to be given in the input inlet.
- Availability of labour for giving daily input.
- Mixing sufficient water to prepare input and collect bio slurry output at regular interval.
- Construction of plastic/straw shed over bio slurry collection chamber.

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

- PC1.** Select suitable location based on factors such as avoidance of water inundation/marshy land/tall tree, availability of cow dung in nearby areas, maximum 200 ft. distance from kitchen point and a minimum 40 ft. distance from pond/river/ tube well etc.
- PC2.** Verify that water is available in the vicinity for making a water solution of dung to be given in the input inlet.
- PC3.** Confirm the availability of labour for giving daily input.
- PC4.** Mix sufficient water to prepare input and also collect bio slurry output at regular interval.
- PC5.** Construct plastic/straw shed over bio slurry collection chamber and keep sufficient space to uplift & dry the collected bio slurry.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Introduction to bio gas.
- KU2.** Difference between Biogas and LPG/CNG.
- KU3.** Advantages of Biogas.
- KU4.** Criteria for selection of site or suitable location Based on factors such as avoidance of water inundation/marshy land/tall tree, availability of cow dung in nearby areas, maximum 200 ft. distance from kitchen point and a minimum 40 ft. distance from pond/river/ tube well etc.
- KU5.** Availability of water in the vicinity for making a water solution of dung to be given in the input inlet.
- KU6.** Availability of labour for giving daily input.
- KU7.** Procedure of mixing sufficient water to prepare input.
- KU8.** Procedure for collection of bio slurry output at regular interval.
- KU9.** Methods of construction of shed over bio slurry collection chamber.



Qualification Pack

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** Demonstrate Biogas generation process with the chart/videos.
- GS2.** Select suitable location based on factors such as avoidance of water inundation/marshy land/tall tree.
- GS3.** Check the availability of cow dung in nearby areas, maximum 200 ft. distance from kitchen point and a minimum 40 ft. distance from pond/river/ tube well etc.
- GS4.** Verify that water is available in the vicinity for making a water solution of dung to be given in the input inlet.
- GS5.** Confirm the availability of labour for giving daily input.
- GS6.** Mix sufficient water to prepare input and also collect bio slurry output at regular interval.
- GS7.** Construct plastic/straw shed over bio slurry collection chamber and keep sufficient space to uplift & dry the collected bio slurry.



Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	20	60	-	-
PC1. Select suitable location based on factors such as avoidance of water inundation/marshy land/tall tree, availability of cow dung in nearby areas, maximum 200 ft. distance from kitchen point and a minimum 40 ft. distance from pond/river/ tube well etc.	-	-	-	-
PC2. Verify that water is available in the vicinity for making a water solution of dung to be given in the input inlet.	-	-	-	-
PC3. Confirm the availability of labour for giving daily input.	-	-	-	-
PC4. Mix sufficient water to prepare input and also collect bio slurry output at regular interval.	-	-	-	-
PC5. Construct plastic/straw shed over bio slurry collection chamber and keep sufficient space to uplift & dry the collected bio slurry.	-	-	-	-
NOS Total	20	60	-	-



Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	WBSC/SGJ/N0204
NOS Name	Identify appropriate site for installing the BGP.
Sector	Green Jobs
Sub-Sector	
Occupation	Bio Gas Plant & Bio Slurry Technician
NSQF Level	3
Credits	1
Version	1.0
Last Reviewed Date	14/12/2023
Next Review Date	14/12/2026
NSQC Clearance Date	14/12/2023



Qualification Pack

WBSC/SGJ/N0205: Construct a Bio Gas Plant with proper care and safety.

Description

Construct a Bio Gas Plant with proper care and safety.

Scope

The scope covers the following :

- Fixed dome Dinabandhu Model family size Bio Gas Plant construction.
- The construction technology and materials as per soil and climatic condition of the State.
- Help of proper supervisory technical expert for BGP and trained masonry support.
- Testing of soil test to find the type of construction.
- Finalization of area to construct the BGP.
- Requirement of construction materials which is required for different types of BGP for different agro-climate.
- Necessary precautions on fire and safety.
- Storing / disposal of dangerous goods and substances.
- Site evacuation procedures according to site policy.
- Personal Protective Equipment (PPE) and their usage.
- Basic first aid and their usage under different circumstances.
- Different fire extinguisher and their usage as per requirement.

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

- PC1.** Plan fixed dome Dinabandhu Model family size Bio Gas Plant construction.
- PC2.** Apply the construction technology and materials as per soil and climatic condition of the State.
- PC3.** Take help of proper supervisory technical expert for BGP and trained masonry support which is essential.
- PC4.** Demonstrate soil test to find the type of construction.
- PC5.** Finalize the area of minimum 150 sq. ft. is required to construct the BGP.
- PC6.** Identify the construction materials which is required as per technical specification for a particular type of BGP for different agro-climatic.
- PC7.** Identify and take necessary precautions on fire and safety.
- PC8.** Identify, handle and store / dispose dangerous goods and substances according to safety regulations and requirements.
- PC9.** Confirm that there is no leakage of gas or don't work with fire in and around BGP.
- PC10.** Identify and observe site evacuation procedures according to site policy.
- PC11.** Identify Personal Protective Equipment (PPE) and use the same as per related working environment.
- PC12.** Identify basic first aid and use them under different circumstances.
- PC13.** Identify different fire extinguisher and use the same as per requirement.



Qualification Pack

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Fixed Dome Dinabandhu Model family size Bio Gas Plant construction.
- KU2.** Construction technology and materials required as per soil and climatic condition of the State.
- KU3.** The components required for the construction of BGP.
- KU4.** The size of Bio Gas plant as per availability of raw materials.
- KU5.** The types of Indian digester as per the design.
- KU6.** The availability of raw materials for the construction at constructions site.
- KU7.** Calculation of the man power required to complete the job.
- KU8.** Instructions from supervisor time to time for installation of gas pipeline which is required technical supervision.
- KU9.** Procedure for excavation work and plant construction work.
- KU10.** Possible risk and hazards in the work environment and exercise safety precautions to minimize injury to self and others.
- KU11.** Relevant safety practices as the work deals with inflatable gas, bleaching water, soap, urea etc. in Bio gas Plant.
- KU12.** Whether there is any leakage of gas or work with fire around BGP.

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** Plan Fixed Dome Dinabandhu Model family size Bio Gas Plant construction.
- GS2.** Apply the construction technology and materials as per soil and climatic condition of the State.
- GS3.** Carry out the soil test to find the type of construction with the help of supervisor.
- GS4.** Mark the construction area for BGP.
- GS5.** Find the construction materials required as per technical specification for a particular type of BGP for different agro-climatic condition.
- GS6.** List and identify the Construction materials which include bricks, sand, stone chips, cement, pipe, iron rod, biogas oven, etc.
- GS7.** Find the man days required to complete the construction, after construction, 15 days is required for curing and then loading of the chamber is to be done with cow dung mixed with water through inlet under loamy & sandy soil.
- GS8.** Find the man days required for construction at hilly areas.
- GS9.** Follow the procedure for installation of gas pipeline which needs technical supervision.
- GS10.** Cure the constructed structure with water for 15 days, after completion of Plant construction.
- GS11.** Give a mixture of cow dung and water 1 kg each through inlet pipeline and the pipeline fittings has to be completed.
- GS12.** Close the valve for seven days and see whether watery cow dung is coming out from the the outlet pipeline. If it comes out, then give 50 kg each of cow dung and water.



Qualification Pack

- GS13.** Check whether gas is coming out through the burner in the kitchen and if found that the gas is not flaming, the full air to be passed and after closing the valve, check on the next day.
- GS14.** Demonstrate Knowledge of Safe working practices on construction sites.
- GS15.** Carry out safe working practices on construction sites.
- GS16.** Identify hazards and procedure to avoid accidents at work sites.
- GS17.** Comply with occupational health and safety requirements relevant to the work.
- GS18.** Assess possible risk and hazards in the work environment and exercise safety precautions to minimize injury to self and others.
- GS19.** Comply relevant safety practices as the work deals with inflatable gas, donot give bleaching water, soap, urea etc. in Bio gas Plant.
- GS20.** Confirm that there is no leakage of gas.
- GS21.** Confirm that there is no work with fire around BGP.



Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	28	210	-	-
PC1. Plan fixed dome Dinabandhu Model family size Bio Gas Plant construction.	-	-	-	-
PC2. Apply the construction technology and materials as per soil and climatic condition of the State.	-	-	-	-
PC3. Take help of proper supervisory technical expert for BGP and trained masonry support which is essential.	-	-	-	-
PC4. Demonstrate soil test to find the type of construction.	-	-	-	-
PC5. Finalize the area of minimum 150 sq. ft. is required to construct the BGP.	-	-	-	-
PC6. Identify the construction materials which is required as per technical specification for a particular type of BGP for different agro-climatic.	-	-	-	-
PC7. Identify and take necessary precautions on fire and safety.	-	-	-	-
PC8. Identify, handle and store / dispose dangerous goods and substances according to safety regulations and requirements.	-	-	-	-
PC9. Confirm that there is no leakage of gas or don't work with fire in and around BGP.	-	-	-	-
PC10. Identify and observe site evacuation procedures according to site policy.	-	-	-	-
PC11. Identify Personal Protective Equipment (PPE) and use the same as per related working environment.	-	-	-	-
PC12. Identify basic first aid and use them under different circumstances.	-	-	-	-
PC13. Identify different fire extinguisher and use the same as per requirement.	-	-	-	-



Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
NOS Total	28	210	-	-



Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	WBSC/SGJ/N0205
NOS Name	Construct a Bio Gas Plant with proper care and safety.
Sector	Green Jobs
Sub-Sector	
Occupation	Bio Gas Plant & Bio Slurry Technician
NSQF Level	3
Credits	3
Version	1.0
Last Reviewed Date	14/12/2023
Next Review Date	14/12/2026
NSQC Clearance Date	14/12/2023



Qualification Pack

WBSC/SGJ/N0206: Install Cooking Apparatus.

Description

Install Cooking Apparatus.

Scope

The scope covers the following :

- Installation of the Gas pipeline, regulator, Gas oven etc.
- Dos & donts of this gas usage.
- Maintenance of the gas line and other apparatus for security reasons.

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

- PC1.** Install the Gas pipeline, regulator, Gas oven etc. Care to be taken to see that no ferrous metal is there in gas line as the Gas contains water vapour and hydrogen sulphide along with methane, which will cause corrosion of the gas line within very short time.
- PC2.** Illustrate dos & don'ts of this gas usage,
- PC3.** Maintain the gas line and other apparatus for security reasons.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** The procedure to connect the gas pipeline and installation of regulator, Gas oven etc.
- KU2.** The safety precaution and care to be taken to see that no ferrous metal is there in gas line as the Gas contains water vapour and hydrogen sulphide along with methane, which will cause corrosion of the gas line within very short time.
- KU3.** Procedure for regular maintenance of the gas line and other apparatus for security reasons.

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** Install Gas pipeline, regulator, Gas oven etc. Care to be taken to see that no ferrous metal is there in gas line as the Gas contains water vapour and hydrogen sulphide along with methane, which will cause corrosion of the gas line within very short time.
- GS2.** Apply dos & donts of this gas usage.
- GS3.** Regularly maintain the gas line and other apparatus for security reasons.



Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	16	40	-	-
PC1. Install the Gas pipeline, regulator, Gas oven etc. Care to be taken to see that no ferrous metal is there in gas line as the Gas contains water vapour and hydrogen sulphide along with methane, which will cause corrosion of the gas line within very short time.	-	-	-	-
PC2. Illustrate dos & don'ts of this gas usage,	-	-	-	-
PC3. Maintain the gas line and other apparatus for security reasons.	-	-	-	-
NOS Total	16	40	-	-



Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	WBSC/SGJ/N0206
NOS Name	Install Cooking Apparatus.
Sector	Green Jobs
Sub-Sector	
Occupation	Bio Gas Plant & Bio Slurry Technician
NSQF Level	3
Credits	1
Version	1.0
Last Reviewed Date	14/12/2023
Next Review Date	14/12/2026
NSQC Clearance Date	14/12/2023



Qualification Pack

WBSC/SGJ/N0207: Carry out post BGP construction activities.

Description

Carry out post BGP construction activities.

Scope

The scope covers the following :

- Carbon credit development and avoidance of fertilizer subsidy with use of Bio slurry.
- Usage of methane as a green substitute of LPG, wood, cattle dung and kerosene etc.
- The procedure of the Plant maintenance to be done on every 10th year on a regular basis.

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

- PC1.** Determine of carbon credit development and avoidance of fertilizer subsidy with use of Bio slurry and use of methane as a green substitute of LPG, wood, cattle dung and kerosene etc. are to be worked out for each sub zone, using BGP.
- PC2.** Demonstrate the procedure step by step of the Plant maintenance which will be done on every 10th year on a regular basis.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Determination of carbon credit development and avoidance of fertilizer subsidy with use of Bio slurry and use of methane as a green substitute of LPG, wood, cattle dung and kerosene etc. are to be worked out for each sub zone, using BGP & community participation in doing so.
- KU2.** Procedure for linking dry/liquid Bio slurry of BGP for organic nutrient management of different agricultural and horticultural crops, replacing chemical fertilizers.
- KU3.** Procedure for general maintenance of the Plant needs to be refreshed on every 10th year on a regular basis.

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** Determine carbon credit development and avoidance of fertilizer subsidy with use of Bio slurry and use of methane as a green substitute of LPG, wood, cattle dung and kerosene etc. are to be worked out for each sub zone, using BGP & community participation in doing so.
- GS2.** Link dry/liquid Bio slurry of BGP for organic nutrient management of different agricultural and horticultural crops, replacing chemical fertilizers.
- GS3.** Refresh the plant on every 10th year on a regular basis.
- GS4.** Remove the cow dung from the plant and wash it.



Qualification Pack

- GS5.** Paint the inner wall of the plant with 5 kg cement mixed with water after drying for two days.
- GS6.** Fill the plant with cow dung following the same rule after drying, as followed for the first filling. Do the same in presence of trained technical expert.



Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	24	120	-	-
PC1. Determine of carbon credit development and avoidance of fertilizer subsidy with use of Bio slurry and use of methane as a green substitute of LPG, wood, cattle dung and kerosene etc. are to be worked out for each sub zone, using BGP.	-	-	-	-
PC2. Demonstrate the procedure step by step of the Plant maintenance which will be done on every 10th year on a regular basis.	-	-	-	-
NOS Total	24	120	-	-



Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	WBSC/SGJ/N0207
NOS Name	Carry out post BGP construction activities.
Sector	Green Jobs
Sub-Sector	
Occupation	Bio Gas Plant & Bio Slurry Technician
NSQF Level	3
Credits	2
Version	1.0
Last Reviewed Date	14/12/2023
Next Review Date	14/12/2026
NSQC Clearance Date	14/12/2023



Qualification Pack

WBSC/SGJ/N0208: Demonstrate the procedure of Bio slurry manure making and Marketing of Organically grown crops.

Description

Demonstrate the procedure of Bio slurry manure making and Marketing of Organically grown crops.

Scope

The scope covers the following :

- Collection of the bio slurry produced from the biogas plant as a by-product in the Outlet Chamber.
- The procedure of mixing the liquid bio slurry with chopped pseudo stem of Banana to increase the potash content of the bio slurry manure.
- Collection of the liquid bio slurry at an upper elevation to make it dried under sun shine and air.
- Mixing the liquid bio slurry with vermicomposting manure to enrich the quality.
- The procedure for mixing of dry Bio slurry manure with microbial bio fertilizers and bio pesticides for value addition.
- Usage of the Bio slurry for fishery, as a source of feed.
- The requirement of bio slurry produced for the cultivation of land and production of crops, vegetables, oil-seeds, pulses and fodder.
- The marketing procedure for organic product.
- Calculation of the profit percentage for the selling the product in Organic huts/markets.
- The scope & way of e-commerce in selling the produce to the consumers.

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

- PC1.** Collect the bio slurry produced from the biogas plant as a by-product in the Outlet Chamber.
- PC2.** Demonstrate the procedure of mixing the liquid bio slurry with chopped pseudo stem of Banana to increase the potash content of the bio slurry manure.
- PC3.** Collect the liquid bio slurry at an upper elevation to make it dried under sun shine and air or it may be directly used in the field as liquid manure at already worked out doses with such frequencies, identified for different crops.
- PC4.** Mix the liquid bio slurry with vermicomposting manure to enrich the quality and reduce the dosage of individual manure.
- PC5.** Follow the procedure for dry Bio slurry manure can be mixed with microbial bio fertilizers and bio pesticides for value addition and enriched manure preparation and organic way of crop cultivation.
- PC6.** Use the Bio slurry for fishery, as a source of feed.
- PC7.** Illustrate the requirement of bio slurry produced for the cultivation of land and production of crops, vegetables, oil-seeds, pulses and fodder.
- PC8.** Demonstrate the marketing procedure for organic product.
- PC9.** Calculate the profit percentage for the selling the product in “Organic huts/markets”,
- PC10.** Identify the scope & way of e-commerce in selling the produce to the consumers.



Qualification Pack

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Procedure to be followed to produce bio slurry.
- KU2.** Uses of bio slurry for fishery as a source of feed.
- KU3.** Comparison of the cost of production which will get reduced over conventional production system.
- KU4.** Procedure for selling the organic products to local market/organic hut.
- KU5.** The scope & way of e-commerce in selling the produce to the consumers.

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** Produce the bio slurry from the biogas plant as a by-product, is to be collected in the Outlet Chamber.
- GS2.** Mix the liquid bio slurry thus produced with chopped pseudo stem of Banana to increase the potash content of the bio slurry manure.
- GS3.** Collect or place the liquid bio slurry can either be at an upper elevation to make it dried under sun shine and air or it may be directly used in the field as liquid manure at already worked out doses with such frequencies, identified for different for different crops.
- GS4.** Mix the liquid bio slurry with vermi composting manure to enrich the quality and reduce the dosage of individual manure.
- GS5.** Mix dried Bio slurry manure with microbial bio fertilizers and bio pesticides for value addition and enriched manure preparation and organic way of crop cultivation.
- GS6.** Use the Bio slurry for fishery, as a source of feed.
- GS7.** Calculate the cost of production conventional production system over organic production system of farming.
- GS8.** Estimate the profit for organic farming by selling the product directly to the consumers.
- GS9.** Evaluate the scope & way of e-commerce in selling the produce to the consumers.



Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	16	50	-	-
PC1. Collect the bio slurry produced from the biogas plant as a by-product in the Outlet Chamber.	-	-	-	-
PC2. Demonstrate the procedure of mixing the liquid bio slurry with chopped pseudo stem of Banana to increase the potash content of the bio slurry manure.	-	-	-	-
PC3. Collect the liquid bio slurry at an upper elevation to make it dried under sun shine and air or it may be directly used in the field as liquid manure at already worked out doses with such frequencies, identified for different crops.	-	-	-	-
PC4. Mix the liquid bio slurry with vermicomposting manure to enrich the quality and reduce the dosage of individual manure.	-	-	-	-
PC5. Follow the procedure for dry Bio slurry manure can be mixed with microbial bio fertilizers and bio pesticides for value addition and enriched manure preparation and organic way of crop cultivation.	-	-	-	-
PC6. Use the Bio slurry for fishery, as a source of feed.	-	-	-	-
PC7. Illustrate the requirement of bio slurry produced for the cultivation of land and production of crops, vegetables, oil-seeds, pulses and fodder.	-	-	-	-
PC8. Demonstrate the marketing procedure for organic product.	-	-	-	-
PC9. Calculate the profit percentage for the selling the product in "Organic huts/markets",	-	-	-	-
PC10. Identify the scope & way of e-commerce in selling the produce to the consumers.	-	-	-	-
NOS Total	16	50	-	-



Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	WBSC/SGJ/N0208
NOS Name	Demonstrate the procedure of Bio slurry manure making and Marketing of Organically grown crops.
Sector	Green Jobs
Sub-Sector	
Occupation	Bio Gas Plant & Bio Slurry Technician
NSQF Level	3
Credits	1
Version	1.0
Last Reviewed Date	14/12/2023
Next Review Date	14/12/2026
NSQC Clearance Date	14/12/2023



Qualification Pack

WBSC/SGJ/N0209: Work in real job situation with special emphasis on basic safety and hazards in this domain (Bio Gas Plant & Bio Slurry Technician).

Description

Work in real job situation with special emphasis on basic safety and hazards in this domain.

Scope

The scope covers the following :

- Work in the real job situation.
- Report preparation on the competency developed.

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

- PC1.** Work in the real job situation with special emphasis on basic safety and hazards in this domain.
- PC2.** Prepare report for this component of training of the course to ensure whether competency has been developed.

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** Work in the real job situation with special emphasis on basic safety and hazards in this domain.
- GS2.** Prepare report for this component of training of the course to ensure whether competency has been developed.



Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	-	120	-	-
PC1. Work in the real job situation with special emphasis on basic safety and hazards in this domain.	-	-	-	-
PC2. Prepare report for this component of training of the course to ensure whether competency has been developed.	-	-	-	-
NOS Total	-	120	-	-



Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	WBSC/SGJ/N0209
NOS Name	Work in real job situation with special emphasis on basic safety and hazards in this domain (Bio Gas Plant & Bio Slurry Technician).
Sector	Green Jobs
Sub-Sector	
Occupation	Bio Gas Plant & Bio Slurry Technician
NSQF Level	3
Credits	2
Version	1.0
Last Reviewed Date	14/12/2023
Next Review Date	14/12/2026
NSQC Clearance Date	14/12/2023



Qualification Pack

DGT/VSQ/N0102: Employability Skills (60 Hours)

Description

This unit is about employability skills, Constitutional values, becoming a professional in the 21st Century, digital, financial, and legal literacy, diversity and Inclusion, English and communication skills, customer service, entrepreneurship, and apprenticeship, getting ready for jobs and career development.

Scope

The scope covers the following :

- Introduction to Employability Skills
- Constitutional values - Citizenship
- Becoming a Professional in the 21st Century
- Basic English Skills
- Career Development & Goal Setting
- Communication Skills
- Diversity & Inclusion
- Financial and Legal Literacy
- Essential Digital Skills
- Entrepreneurship
- Customer Service
- Getting ready for Apprenticeship & Jobs

Elements and Performance Criteria

Introduction to Employability Skills

To be competent, the user/individual on the job must be able to:

- PC1.** identify employability skills required for jobs in various industries
- PC2.** identify and explore learning and employability portals

Constitutional values – Citizenship

To be competent, the user/individual on the job must be able to:

- PC3.** recognize the significance of constitutional values, including civic rights and duties, citizenship, responsibility towards society etc. and personal values and ethics such as honesty, integrity, caring and respecting others, etc.
- PC4.** follow environmentally sustainable practices

Becoming a Professional in the 21st Century

To be competent, the user/individual on the job must be able to:

- PC5.** recognize the significance of 21st Century Skills for employment
- PC6.** practice the 21st Century Skills such as Self-Awareness, Behaviour Skills, time management, critical and adaptive thinking, problem-solving, creative thinking, social and cultural awareness, emotional awareness, learning to learn for continuous learning etc. in personal and professional life

Basic English Skills

To be competent, the user/individual on the job must be able to:



Qualification Pack

- PC7.** use basic English for everyday conversation in different contexts, in person and over the telephone
- PC8.** read and understand routine information, notes, instructions, mails, letters etc. written in English
- PC9.** write short messages, notes, letters, e-mails etc. in English

Career Development & Goal Setting

To be competent, the user/individual on the job must be able to:

- PC10.** understand the difference between job and career
- PC11.** prepare a career development plan with short- and long-term goals, based on aptitude

Communication Skills

To be competent, the user/individual on the job must be able to:

- PC12.** follow verbal and non-verbal communication etiquette and active listening techniques in various settings
- PC13.** work collaboratively with others in a team

Diversity & Inclusion

To be competent, the user/individual on the job must be able to:

- PC14.** communicate and behave appropriately with all genders and PwD
- PC15.** escalate any issues related to sexual harassment at workplace according to POSH Act

Financial and Legal Literacy

To be competent, the user/individual on the job must be able to:

- PC16.** select financial institutions, products and services as per requirement
- PC17.** carry out offline and online financial transactions, safely and securely
- PC18.** identify common components of salary and compute income, expenses, taxes, investments etc
- PC19.** identify relevant rights and laws and use legal aids to fight against legal exploitation

Essential Digital Skills

To be competent, the user/individual on the job must be able to:

- PC20.** operate digital devices and carry out basic internet operations securely and safely
- PC21.** use e-mail and social media platforms and virtual collaboration tools to work effectively
- PC22.** use basic features of word processor, spreadsheets, and presentations

Entrepreneurship

To be competent, the user/individual on the job must be able to:

- PC23.** identify different types of Entrepreneurship and Enterprises and assess opportunities for potential business through research
- PC24.** develop a business plan and a work model, considering the 4Ps of Marketing Product, Price, Place and Promotion
- PC25.** identify sources of funding, anticipate, and mitigate any financial/ legal hurdles for the potential business opportunity

Customer Service

To be competent, the user/individual on the job must be able to:

- PC26.** identify different types of customers
- PC27.** identify and respond to customer requests and needs in a professional manner.



Qualification Pack

PC28. follow appropriate hygiene and grooming standards

Getting ready for apprenticeship & Jobs

To be competent, the user/individual on the job must be able to:

PC29. create a professional Curriculum vitae (Résumé)

PC30. search for suitable jobs using reliable offline and online sources such as Employment exchange, recruitment agencies, newspapers etc. and job portals, respectively

PC31. apply to identified job openings using offline /online methods as per requirement

PC32. answer questions politely, with clarity and confidence, during recruitment and selection

PC33. identify apprenticeship opportunities and register for it as per guidelines and requirements

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. need for employability skills and different learning and employability related portals

KU2. various constitutional and personal values

KU3. different environmentally sustainable practices and their importance

KU4. Twenty first (21st) century skills and their importance

KU5. how to use English language for effective verbal (face to face and telephonic) and written communication in formal and informal set up

KU6. importance of career development and setting long- and short-term goals

KU7. about effective communication

KU8. POSH Act

KU9. Gender sensitivity and inclusivity

KU10. different types of financial institutes, products, and services

KU11. how to compute income and expenditure

KU12. importance of maintaining safety and security in offline and online financial transactions

KU13. different legal rights and laws

KU14. different types of digital devices and the procedure to operate them safely and securely

KU15. how to create and operate an e- mail account and use applications such as word processors, spreadsheets etc.

KU16. how to identify business opportunities

KU17. types and needs of customers

KU18. how to apply for a job and prepare for an interview

KU19. apprenticeship scheme and the process of registering on apprenticeship portal

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. read and write different types of documents/instructions/correspondence

GS2. communicate effectively using appropriate language in formal and informal settings



Qualification Pack

- GS3.** behave politely and appropriately with all
- GS4.** how to work in a virtual mode
- GS5.** perform calculations efficiently
- GS6.** solve problems effectively
- GS7.** pay attention to details
- GS8.** manage time efficiently
- GS9.** maintain hygiene and sanitization to avoid infection



Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Introduction to Employability Skills</i>	1	1	-	-
PC1. identify employability skills required for jobs in various industries	-	-	-	-
PC2. identify and explore learning and employability portals	-	-	-	-
<i>Constitutional values – Citizenship</i>	1	1	-	-
PC3. recognize the significance of constitutional values, including civic rights and duties, citizenship, responsibility towards society etc. and personal values and ethics such as honesty, integrity, caring and respecting others, etc.	-	-	-	-
PC4. follow environmentally sustainable practices	-	-	-	-
<i>Becoming a Professional in the 21st Century</i>	2	4	-	-
PC5. recognize the significance of 21st Century Skills for employment	-	-	-	-
PC6. practice the 21st Century Skills such as Self-Awareness, Behaviour Skills, time management, critical and adaptive thinking, problem-solving, creative thinking, social and cultural awareness, emotional awareness, learning to learn for continuous learning etc. in personal and professional life	-	-	-	-
<i>Basic English Skills</i>	2	3	-	-
PC7. use basic English for everyday conversation in different contexts, in person and over the telephone	-	-	-	-
PC8. read and understand routine information, notes, instructions, mails, letters etc. written in English	-	-	-	-
PC9. write short messages, notes, letters, e-mails etc. in English	-	-	-	-
<i>Career Development & Goal Setting</i>	1	2	-	-



Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC10. understand the difference between job and career	-	-	-	-
PC11. prepare a career development plan with short- and long-term goals, based on aptitude	-	-	-	-
<i>Communication Skills</i>	2	2	-	-
PC12. follow verbal and non-verbal communication etiquette and active listening techniques in various settings	-	-	-	-
PC13. work collaboratively with others in a team	-	-	-	-
<i>Diversity & Inclusion</i>	1	2	-	-
PC14. communicate and behave appropriately with all genders and PwD	-	-	-	-
PC15. escalate any issues related to sexual harassment at workplace according to POSH Act	-	-	-	-
<i>Financial and Legal Literacy</i>	2	3	-	-
PC16. select financial institutions, products and services as per requirement	-	-	-	-
PC17. carry out offline and online financial transactions, safely and securely	-	-	-	-
PC18. identify common components of salary and compute income, expenses, taxes, investments etc	-	-	-	-
PC19. identify relevant rights and laws and use legal aids to fight against legal exploitation	-	-	-	-
<i>Essential Digital Skills</i>	3	4	-	-
PC20. operate digital devices and carry out basic internet operations securely and safely	-	-	-	-
PC21. use e- mail and social media platforms and virtual collaboration tools to work effectively	-	-	-	-
PC22. use basic features of word processor, spreadsheets, and presentations	-	-	-	-



Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Entrepreneurship</i>	2	3	-	-
PC23. identify different types of Entrepreneurship and Enterprises and assess opportunities for potential business through research	-	-	-	-
PC24. develop a business plan and a work model, considering the 4Ps of Marketing Product, Price, Place and Promotion	-	-	-	-
PC25. identify sources of funding, anticipate, and mitigate any financial/ legal hurdles for the potential business opportunity	-	-	-	-
<i>Customer Service</i>	1	2	-	-
PC26. identify different types of customers	-	-	-	-
PC27. identify and respond to customer requests and needs in a professional manner.	-	-	-	-
PC28. follow appropriate hygiene and grooming standards	-	-	-	-
<i>Getting ready for apprenticeship & Jobs</i>	2	3	-	-
PC29. create a professional Curriculum vitae (Résumé)	-	-	-	-
PC30. search for suitable jobs using reliable offline and online sources such as Employment exchange, recruitment agencies, newspapers etc. and job portals, respectively	-	-	-	-
PC31. apply to identified job openings using offline /online methods as per requirement	-	-	-	-
PC32. answer questions politely, with clarity and confidence, during recruitment and selection	-	-	-	-
PC33. identify apprenticeship opportunities and register for it as per guidelines and requirements	-	-	-	-
NOS Total	20	30	-	-



Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	DGT/VSQ/N0102
NOS Name	Employability Skills (60 Hours)
Sector	Cross Sectoral
Sub-Sector	Professional Skills
Occupation	Employability
NSQF Level	4
Credits	2
Version	1.0
Last Reviewed Date	18/02/2025
Next Review Date	18/02/2028
NSQF Clearance Date	18/02/2025

Assessment Guidelines and Assessment Weightage

Assessment Guidelines

Assessment will be based on the concept of Independent Assessors empaneled with West Bengal State Council of Technical & Vocational Education & Skill Development (WBSCT&VE&SD), identified, selected, trained and certified on Assessment techniques. These Assessors would be aligned to assess as per the laid down criteria.

WBSCT&VE&SD would conduct assessment only at the training centers or designated testing centers authorized by WBSCT&VE&SD. Ideally, the assessment will be a continuous process comprising of two distinct steps:

- Continuous assessment by Trainers
- Term end /Final Assessment by WBSCT&VE&SD

Each National Occupational Standard (NOS) in the respective QPs will be assigned weightage. Each Performance Criteria in the NOS will be assigned marks for theory and/or practical based on relative importance and criticality of function. This will facilitate preparation of question bank / paper sets for each of the QPs. Each of these papers sets/question banks created by subject matter experts through WBSCT&VE&SD, especially with regard to the practical test and the defined tolerances, finish, accuracy



Qualification Pack

etc. The following tools are proposed to be used for final assessment:

i. Written Test: This will comprise of (i) True/False Statements and/or (ii) Multiple Choice Questions and/or (iii) Matching Type Questions. Online system for this will be preferred.

ii. Practical Test: This will comprise a test job to be prepared as per project briefing following appropriate working steps, using necessary tools, equipment and instruments. Through observation it will be possible to ascertain candidate's aptitude, attention to details, quality consciousness etc.

iii. Structured Viva-voce: This tool will be used to assess the conceptual understanding and the behavioral aspects as regards the job role and the specific task at hand.

Minimum Aggregate Passing % at QP Level : 70

(Please note: Every Trainee should score a minimum aggregate passing percentage as specified above, to successfully clear the Qualification Pack assessment.)

Assessment Weightage

Compulsory NOS

National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
WBSC/SGJ/N0201. Identify Biomass resource and Bio-energy potential.	10	80	-	-	90	9
WBSC/SGJ/N0202. Illustrate mechanism of Biogas Systems.	16	60	-	-	76	8
WBSC/SGJ/N0203. Explain the process of Biomass Gasifiers.	20	60	-	-	80	8
WBSC/SGJ/N0204. Identify appropriate site for installing the BGP.	20	60	-	-	80	8
WBSC/SGJ/N0205. Construct a Bio Gas Plant with proper care and safety.	28	210	-	-	238	24
WBSC/SGJ/N0206. Install Cooking Apparatus.	16	40	-	-	56	6
WBSC/SGJ/N0207. Carry out post BGP construction activities.	24	120	-	-	144	14



Qualification Pack

National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
WBSC/SGJ/N0208.Demonstrate the procedure of Bio slurry manure making and Marketing of Organically grown crops.	16	50	-	-	66	6
WBSC/SGJ/N0209.Work in real job situation with special emphasis on basic safety and hazards in this domain (Bio Gas Plant & Bio Slurry Technician).	-	120	-	-	120	12
DGT/VSQ/N0102.Employability Skills (60 Hours)	20	30	-	-	50	5
Total	170	830	-	-	1000	100



Qualification Pack

Acronyms

NOS	National Occupational Standard(s)
NSQF	National Skills Qualifications Framework
QP	Qualifications Pack
TVET	Technical and Vocational Education and Training
NOS	National Occupational Standard(s)
NSQF	National Skills Qualifications Framework
QP	Qualifications Pack
TVET	Technical and Vocational Education and Training
AA	Assessment Agency
AB	Awarding Body
ISCO	International Standard Classification of Occupations
NCO	National Classification of Occupations
NCrF	National Credit Framework
NQR	National Qualification Register



Qualification Pack

Glossary

Sector	Sector is a conglomeration of different business operations having similar business and interests. It may also be defined as a distinct subset of the economy whose components share similar characteristics and interests.
Sub-sector	Sub-sector is derived from a further breakdown based on the characteristics and interests of its components.
Occupation	Occupation is a set of job roles, which perform similar/ related set of functions in an industry.
Job role	Job role defines a unique set of functions that together form a unique employment opportunity in an organisation.
Occupational Standards (OS)	OS specify the standards of performance an individual must achieve when carrying out a function in the workplace, together with the Knowledge and Understanding (KU) they need to meet that standard consistently. Occupational Standards are applicable both in the Indian and global contexts.
Performance Criteria (PC)	Performance Criteria (PC) are statements that together specify the standard of performance required when carrying out a task.
National Occupational Standards (NOS)	NOS are occupational standards which apply uniquely in the Indian context.
Qualifications Pack (QP)	QP comprises the set of OS, together with the educational, training and other criteria required to perform a job role. A QP is assigned a unique qualifications pack code.
Unit Code	Unit code is a unique identifier for an Occupational Standard, which is denoted by an 'N'
Unit Title	Unit title gives a clear overall statement about what the incumbent should be able to do.
Description	Description gives a short summary of the unit content. This would be helpful to anyone searching on a database to verify that this is the appropriate OS they are looking for.
Scope	Scope is a set of statements specifying the range of variables that an individual may have to deal with in carrying out the function which have a critical impact on quality of performance required.
Knowledge and Understanding (KU)	Knowledge and Understanding (KU) are statements which together specify the technical, generic, professional and organisational specific knowledge that an individual needs in order to perform to the required standard.



Qualification Pack

Organisational Context	Organisational context includes the way the organisation is structured and how it operates, including the extent of operative knowledge managers have of their relevant areas of responsibility.
Technical Knowledge	Technical knowledge is the specific knowledge needed to accomplish specific designated responsibilities.
Core Skills/ Generic Skills (GS)	Core skills or Generic Skills (GS) are a group of skills that are the key to learning and working in today's world. These skills are typically needed in any work environment in today's world. These skills are typically needed in any work environment. In the context of the OS, these include communication related skills that are applicable to most job roles.
Electives	Electives are NOS/set of NOS that are identified by the sector as contributive to specialization in a job role. There may be multiple electives within a QP for each specialized job role. Trainees must select at least one elective for the successful completion of a QP with Electives.
Options	Options are NOS/set of NOS that are identified by the sector as additional skills. There may be multiple options within a QP. It is not mandatory to select any of the options to complete a QP with Options.
Sector	Sector is a conglomeration of different business operations having similar business and interests. It may also be defined as a distinct subset of the economy whose components share similar characteristics and interests.
Sub-sector	Sub-sector is derived from a further breakdown based on the characteristics and interests of its components.
Occupation	Occupation is a set of job roles, which perform similar/ related set of functions in an industry.
Job role	Job role defines a unique set of functions that together form a unique employment opportunity in an organisation.
Occupational Standards (OS)	OS specify the standards of performance an individual must achieve when carrying out a function in the workplace, together with the Knowledge and Understanding (KU) they need to meet that standard consistently. Occupational Standards are applicable both in the Indian and global contexts.
Performance Criteria (PC)	Performance Criteria (PC) are statements that together specify the standard of performance required when carrying out a task.
National Occupational Standard	National Occupational Standards (NOS) are occupational standards which apply uniquely in the Indian context.
Qualifications Pack (QP)	QP comprises the set of OS, together with the educational, training and other criteria required to perform a job role. A QP is assigned a unique qualifications pack code.



Qualification Pack

Unit Code	Unit code is a unique identifier for an Occupational Standard, which is denoted by an N
Unit Title	Unit title gives a clear overall statement about what the incumbent should be able to do.
Description	Description gives a short summary of the unit content. This would be helpful to anyone searching on a database to verify that this is the appropriate OS they are looking for.
Scope	Scope is a set of statements specifying the range of variables that an individual may have to deal with in carrying out the function which have a critical impact on quality of performance required.
Knowledge and Understanding	Knowledge and Understanding (KU) are statements which together specify the technical, generic, professional and organisational specific knowledge that an individual needs in order to perform to the required standard.
Organisational Context	Organisational context includes the way the organisation is structured and how it operates, including the extent of operative knowledge managers have of their relevant areas of responsibility.
Technical Knowledge	Technical knowledge is the specific knowledge needed to accomplish specific designated responsibilities.
Core Skills/ Generic Skills	Core skills or Generic Skills (GS) are a group of skills that are the key to learning and working in today's world. These skills are typically needed in any work environment in today's world. These skills are typically needed in any work environment. In the context of the OS, these include communication related skills that are applicable to most job roles.
Electives	Electives are NOS/set of NOS that are identified by the sector as contributive to specialization in a job role. There may be multiple electives within a QP for each specialized job role. Trainees must select at least one elective for the successful completion of a QP with Electives.
Options	Options are NOS/set of NOS that are identified by the sector as additional skills. There may be multiple options within a QP. It is not mandatory to select any of the options to complete a QP with Options.
Qualification	A formal outcome of an assessment and validation process which is obtained when a competent body determines that an individual has achieved learning outcomes to given standards.
Qualification File	A Qualification File is a template designed to capture necessary information of a Qualification from the perspective of NSQF compliance. The Qualification File will be normally submitted by the awarding body for the qualification.
Long Term Training	Long-term skilling means any vocational training program undertaken for a year and above. https://ncvet.gov.in/sites/default/files/NCVET.pdf .